

Lucas Soriano

Python | AI | Automation Engineer

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PROFESSIONAL SUMMARY

Senior AI and Automation Engineer with 8+ years of experience designing and deploying production-grade generative AI pipelines and scalable backend systems. Specialized in Python-driven RAG architectures, vector databases, and multi-agent workflows using LangChain and LangGraph, improving information retrieval accuracy by 45%. Proven expertise in cloud-native deployments on AWS with CI/CD automation, integrating AI services into enterprise platforms.

TECHNICAL SKILLS

- **AI and Automation:** LLMs, OpenAI API, RAG Systems, AI Agents, Prompt Engineering, AI Workflow Automation, Vector Databases, Embeddings, Generative AI, Reranking Mechanisms, LLM APIs Gemini, LLM APIs Anthropic, LLM APIs Llama, LLM APIs Ollama, GitHub Copilot, Generative AI - GANs, Generative AI - Diffusion Models, Retrieval Augmented Generation (RAG) Pipelines
- **AI Frameworks:** LangChain, LangGraph, LlamaIndex, Hugging Face, PyTorch
- **Programming:** Python, SQL, JavaScript, TypeScript, Pandas, NumPy, Scikit-learn
- **Backend & APIs:** FastAPI, Django, Flask, REST APIs, GraphQL, Microservices
- **Data & Databases:** PostgreSQL, MongoDB, Redis, FAISS, Pinecone, Data Ingestion And Processing Pipelines
- **Cloud & DevOps:** AWS, Docker, Kubernetes, CI/CD, Linux, Git, Azure, GCP

PROFESSIONAL EXPERIENCE

REALTECH Inc

Oct 2022 - Apr 2026

Senior AI & Automation Engineer

remote

- Architected and developed generative AI-powered software solutions by integrating LLMs, robust data ingestion and processing pipelines, and scalable backend systems to solve complex business challenges and improve operational efficiency.
- Built production-ready AI applications using Python, OpenAI, Gemini, Anthropic, Llama, and Ollama APIs, RAG pipelines, embeddings, vector databases, and advanced rerankers leveraging Pandas, NumPy, and Scikit-learn to improve information retrieval accuracy by 45% and enable faster access to enterprise knowledge.
- Designed and implemented AI agent workflows using tool calling, external API integrations, and automated decision processes to streamline multi-step business operations and reduce manual workload by 35%.
- Developed intelligent automation solutions using Python scripts, workflow automation platforms, and system integrations to eliminate repetitive processes and improve productivity across business operations.
- Engineered scalable backend services using FastAPI, Django, PostgreSQL, Redis, Docker, and deployed on AWS, Azure, and GCP cloud platforms while optimizing AI application performance, reliability, and maintainability.
- Collaborated with product teams, engineers, and stakeholders to identify AI opportunities, transform business requirements into practical solutions, integrate AI-generated code using GitHub Copilot for rapid prototyping, and deliver production-ready AI systems.
- Implemented generative AI - GANs and diffusion models for synthetic data generation to augment training datasets, improving model accuracy and diversity.
- Built and maintained CI/CD pipelines using GitHub Actions and Docker for deploying and updating generative AI services and pipelines, reducing deployment time by 40%.
- Orchestrated data flow between various AI services, databases, and backend systems within the generative AI context to ensure seamless integration and real-time processing.

Avenue Code

Oct 2020 - Aug 2022

Senior Python Engineer

remote

- Engineered and optimized Python Django and FastAPI backends, introducing modular architectures, async data pipelines, and caching layers that reduced API latency by 35% and enabled seamless scaling for millions of global users.
- Built distributed backend services and asynchronous processing workflows to handle complex business operations, background tasks, and large-scale data processing requirements.
- Automated deployments with Docker, Kubernetes, and GitHub Actions, achieving zero-downtime releases and reducing release cycles from bi-weekly to daily.
- Increased reliability through extensive unit, integration, and end-to-end testing, reaching 90% coverage and reducing production incidents by 40%.
- Provided unit, integration and E2E tests using pytest and Testcontainers.
- Mentored engineers by conducting code reviews, technical workshops, and process improvements, elevating team-wide productivity and code quality.

DB1 Global Software

Jul 2019 - Sep 2020

Senior Python Developer

hybrid

- Designed and implemented Python Django and Flask applications with clean modular architectures, service layers, and dependency injection, improving long-term maintainability and onboarding speed by 40%.
- Developed secure and scalable REST and GraphQL APIs, integrating OAuth2.0 and JWT authentication, audit logging, and role-based permissions for enterprise clients.
- Optimized PostgreSQL-heavy systems by redesigning schemas, applying indexing, and adding Redis caching, which cut query times from seconds to milliseconds and powered real-time analytics dashboards.
- Worked directly with the Product Owner to design solutions leveraging available AWS Services support scaling up the application designed to use a multi-account AWS environment.
- Delivered mobile-first experiences by integrating Python backends with React and Angular frontends, improving end-user engagement by 20%.
- Improved reliability through automated testing frameworks (PyTest, Cypress), achieving 85%+ coverage and lowering production incidents by 30%.

BRQ Digital Solutions

May 2018 - Jun 2019

Software Engineer

onsite

- Assisted with backend development using Python, PHP, JavaScript, and SQL during Internship period.
- Supported backend development activities including API implementation, database operations, and application maintenance.
- Developed small automation scripts and technical utilities to support development workflows and reduce repetitive manual tasks.
- Participated in testing activities, troubleshooting sessions, and code improvement efforts to increase application quality and reliability.

EDUCATION

University of São Paulo(USP)

Mar 2014 - Apr 2018

Bachelor's Degree, Computer Science